Predicting Corporate Bankruptcy Using Tree-Based Models and Ensemble Learning

MGMT 590 (ML) Final Project

Team MoneyMakers

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Introduction

Project Fundamentals

"For this project, we wanted to use machine learning to predict modern stock volatility in times of economic contraction. Utilizing modern financial data and metrics to measure accuracy"

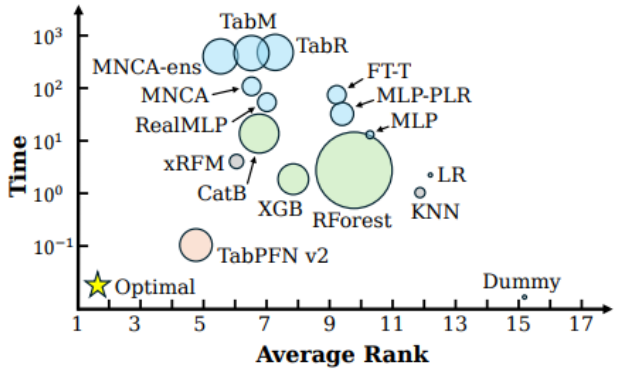
Research

Article One

Ye, H. J., Liu, S. Y., Cai, H. R., Zhou, Q. L., & Zhan, D. C. (2024). A Closer Look at Deep Learning Methods on Tabular Datasets.

Evaluation of a plethora of machine learning methods on a wide variety of data sets(300 common size datasets of different industries). Are certain methods better used on certain forms of data, or is there a clear winner?

- Deep learning becomes relevant when learning complex relations within data and performs well on larger data sets.
- Tree based methods perform remarkably well with most types of data and excel with small-medium sized data sets.
- Pre trained models such as TabFPN are top performers, but can only be applied to small-medium data sets(10,000 rows or less).



(d) All Tasks

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Research

Article Two

Y., Luo, Y., Wang, Z., & Zhang, X. (2026). Uncertainty-Adjusted Sorting for Asset Pricing with Machine Learning.

Implementation of generative artificial intelligence on tabular financial data with an emphasis on asset return prediction. Using these methods to optimize portfolio production, and use intervals instead of point prediction.

- There is massive amounts of uncertainty within financial data. Statistical methods stress uncertainty but point predictions are still used. Using generative models to predict intervals of returns can lead to an increase in sharpe-ratio outcomes.
- The main problem with financial data is too much noise to find a single signal (signal to noise ratio)
- Minimizing error is key in predicting returns on stocks. There is no point in the ML model if the prediction cannot be trusted.

Research

Article Three

Kabir, M. R., Bhadra, D., Ridoy, M., & Milanova, M. (2025). LSTM–Transformer-Based Robust Hybrid Deep Learning Model for Financial Time Series Forecasting.

RNNs and LSTMS models have been recently utilized and proven good models for financial data. They handle the mass robustness and non-linearity very well compared to other models. The paper implemented a similar process below to tackle the problem of financial data:

- **LSTM Layer:** Used initially to capture long-duration context and provide robustness against noise or missing data.
- **Modified Transformer (mTrans):** Takes the feature vectors from the LSTM and uses a self-attention mechanism to weigh relationships between different parts of the data.
- **Multilayer Perceptron (MLP):** Serves as the final layer to output the predicted ending values of the stock data.

Initial Model Building Part One

Data Preprocessing and Model Building

Applying our foundational knowledge of machine learning, content learned in class, and literature review we wanted to test three models:

- Boosted tree model
- Recurrent neural network model
- Ensemble neural network model(Inspired from third research paper)

We implemented modern data, only using 2017-2019 for the train split and 2020 for the test. This highlights the COVID market crash, clearly testing if our model can handle unforeseen volatility.

Train/Test Data Structure:

```
Shape of X_train: (123139, 177)
Shape of X_test: (36104, 177)
Shape of y_train: (123139,)
Shape of y_test: (36104,)
```

Why use all variables:

We felt that using the entire variable range in the scope of our problem was appropriate. Volatility effects all markets overtime and focusing on one area might produce bias within our study.

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Initial Model Building Part Two

Accuracy Metrics and Time Series Measure

A serious problem withing dealing with financial data is look ahead bias. Machine Learning models will prefer future data over present.

Do we have an issue with look ahead bias?

No, due to all variable being within specific months, this addresses the look ahead bias. The model will consider variables within only one month at a time.

Metrics used to evaluate model performance with volatility:

- Mean squared error and Mean absolute error
  - What is the average error in terms of excess return
- Directional Accuracy(Hit Ratio)
  - How often do we predict the sign of the excess return correctly?

Data Processing Notes

Imputation Methods

How should we impute?

- Using mean would include different industries performance, influencing stocks that have missing data.
- We don’t use median for numerical for many reasons, the largest of which is the issue we would have with our data being time-series.
- KNN is not useful here because when it searches for the most similar data it may impute data from a different year which we don’t want.

How did we impute?

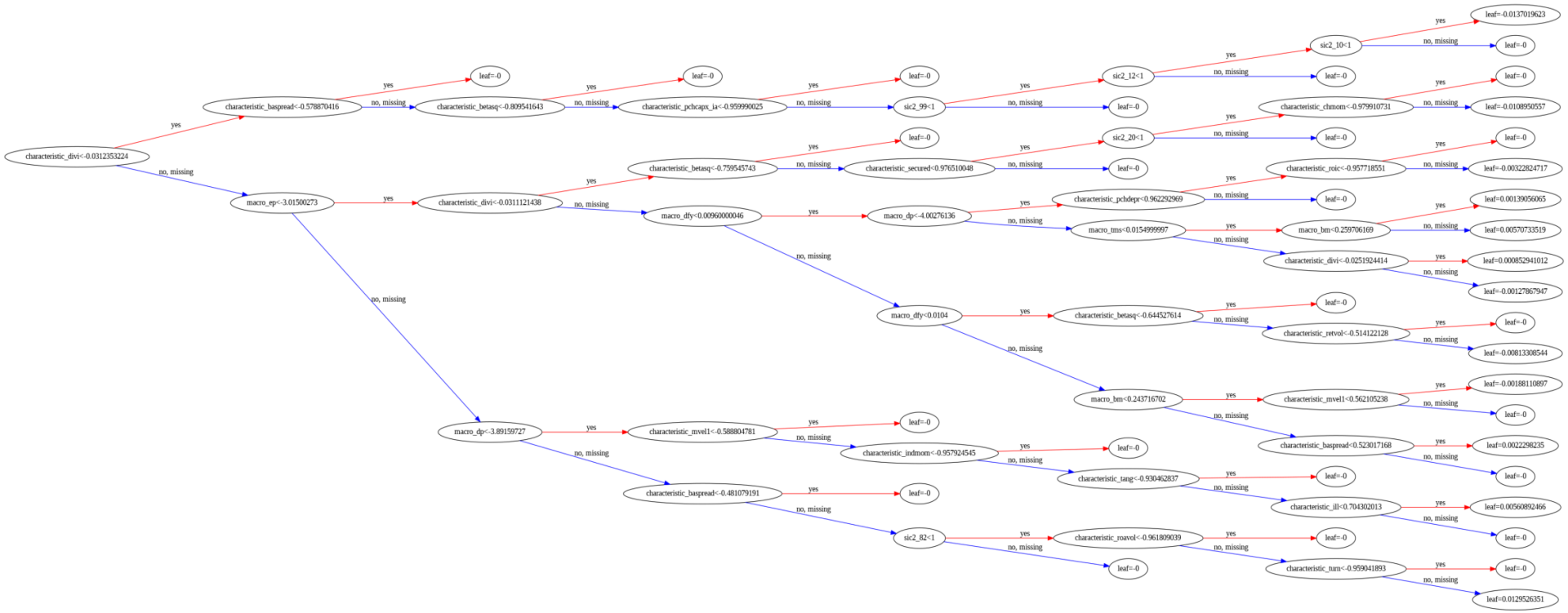
Our group decided to use forward-fill, sorting by stock ID (Permno) and only using that stocks previous performance as a replacement. While there is the downside that this method will not account for an increase or decrease in performance, this was the most successful method we implemented and it significantly improved our scores.

XGBoost Model

Research and Optimization

As we previously saw in our research, XGBoost typically performs well while not being computationally intensive. This was evident as it outperformed our basic RNN. Unlike the basic RNN, our XGBoost model faced heavy hyperparameter optimization through the library RandomizedSearchCV.

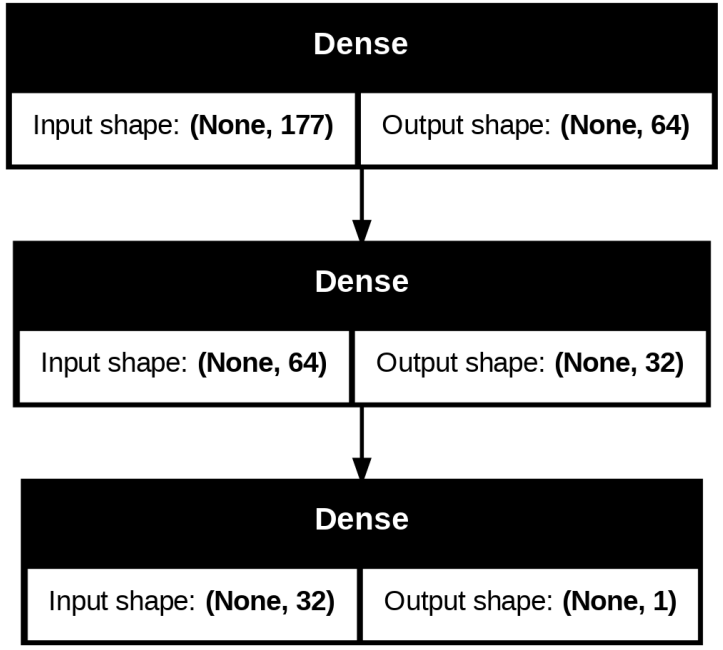
Overall Performance
Hit Ratio: 0.5404
MSE: 0.0769



Deep Neural Network Model

Simple RNN Model

Our baseline deep learning model, implementing code from class homeworks. Simple structure with three layers overall.
The structure is a simple deep neural network model that is in the 'feed-forward' format. Relu used for the activation function.
Overall Performance of the model:
Hit Ratio: 0.5318
MSE: 0.0822
MAE: 0.1634



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Hybrid LSTM-Transformer Model

Ensemble Methods

```
Test Loss (MSE): 0.0787
Test MAE: 0.1416
```

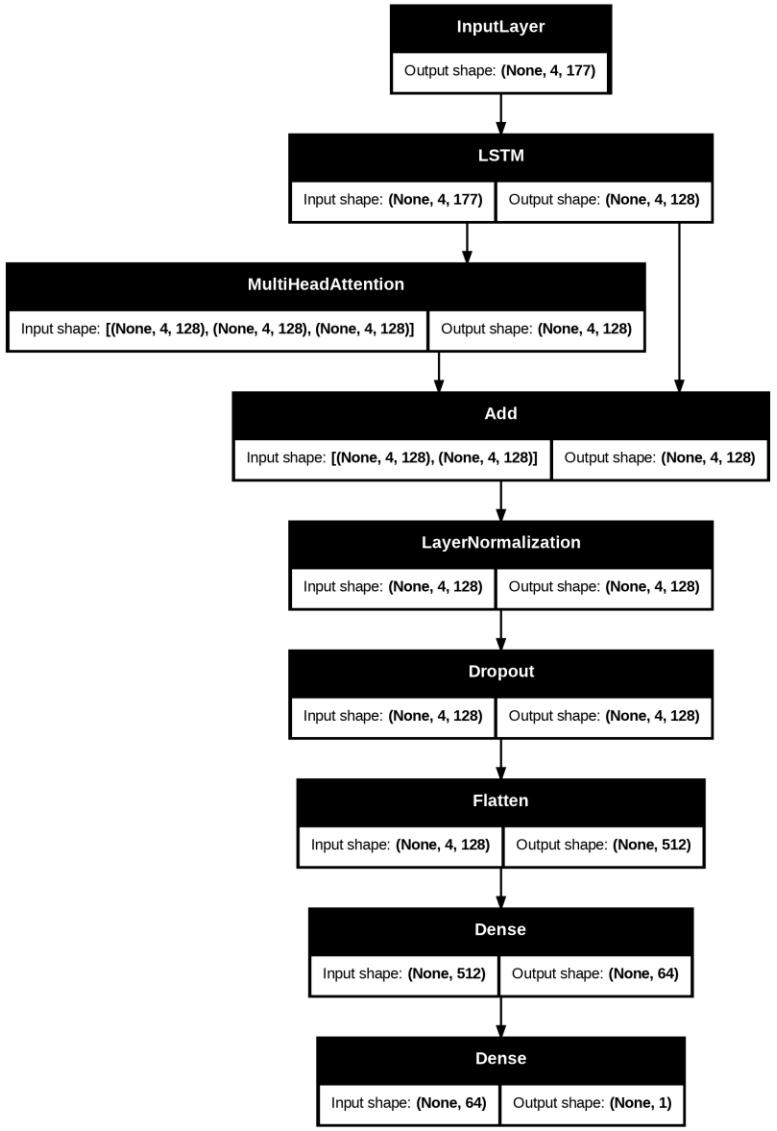
This was a model derived from research, to implement LSTM to measure time series . The model follows a rough structure of:

We set a time window, meaning the model looks back on data to make decisions on how the market is going to move. It processes this information and tries to interpret the volatility in the market (will the market be more stable in four months compared to now)?

The mulithead attention considers all data regardless of the time to understand the data. It is multi to have different viewpoints into certain aspects of the data (well rounded opinion on forecast of data).

The attention layer can be too 'aggressive' and loose signals from the previous layers. The normalization ensures there is no 'forgetting' these valuable insights.

Finally it is fed into a multi-layer perceptron for the predicted excess return value. This is done through condensing dense layers.



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Insights

Key Takeaways

- Signal vs Noise
  - Financial data is very noisy and hard to find the signal(low error means receiving the signal)
  - Exogenous Factors (many factors effect stock prices on the daily).
- What determines a good hit ratio?
  - We felt that exceeding 50% signaled good results from the model. Financial data is very hard to predict and though this seems low, it is better than random guessing or all positive returns.
- Tree based models are very good at complex non-linear data.
  - We were surprised with how it handled this complex large data set. This further reinforces the research around xgboost and boosted tree methods.
- Data Quality
  - The imputation of missing values drastically changed our evaluation metrics. Data quality has a massive impact on the outcomes of any machine learning model.

Insights Pt2

Key Takeaways

The LSTM model revealed how deep learning models can truly find the signals in complex data such as this.
How does the time step/window length effect the performance of the model?
Based on the values we tested, the hit ratio varied from 0.54-0.81.
This interval may seem like a good show of accuracy, yet is misleading.
Tweaking the window shows what months will be reserved as 'reference material' for the model in the test set.
For example:
Set time step to four months. This means the first four months of the month in 2020 will be used as reference to gather insight into the noise. If we increase this it generalizes more but also leaves out the volatile period we wanted to predict. So it is important to select an appropriate window length or results will be unmeaningful.

We decided that a potential time step would be **2 months**, since the recession in 2020 was quite short. If longer the model will realize the market has recovered and predict positive excess returns.

Overall the model was incredibly interesting and insightful. It was really fun to learn a complex application of the models we learned in class.

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Thank You

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